

Peter VanNostrand

Curriculum Vitae

Data Science PhD Candidate
Worcester Polytechnic Institute

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Research Interests

Explainable AI, Algorithmic Bias, Human-Centric Computing, Responsible Data Science Education

Education

Worcester Polytechnic Institute , Worcester MA PhD in Data Science, GPA 4.0/4.0	Aug 2020 – Present
Worcester Polytechnic Institute , Worcester MA MS in Data Science, GPA 4.0/4.0	Aug 2020 – Dec 2022
University at Buffalo (SUNY Buffalo) , Buffalo NY BS in Computer Engineering, GPA: 3.94/4.0 BS in Electrical Engineering, GPA: 3.94/4.0	Aug 2016 – May 2020

Publications

1. **Generating the Unknown: Embedding-Based Anomaly Generation at the Boundary of Normal.** *Dennis M. Hofmann, Joshua C. DeOliveira, Peter M. VanNostrand, Lei Ma, Lei Cao, and Elke Rundensteiner.* (SIGMOD 2027, Under Review).
2. **Options Without Agency: Diversity as a Requirement for Fair Actionable Recourse.** *Peter M. VanNostrand, Dennis M. Hofmann, Lei Ma, and Elke A. Rundensteiner.* Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT 2026, Accepted to Publish).
3. **Detect, Localize, and Explain: Interactive Hierarchical Log Anomaly Analytics with LLM Augmentation.** *Lei Ma, Suhani Chaudhary, Ethan Shanbaum, Athanasios Tassiadamis, Peter M. VanNostrand, Dennis M. Hofmann, Haowen Xu, and Elke Rundensteiner.* (VLDB 2026, Under Review).
4. **KRONE: Hierarchical and Modular Log Anomaly Detection.** *Lei Ma, Jinyang Liu, Tieying Zhang, Peter M. VanNostrand, Dennis M. Hofmann, Lei Cao, Elke A. Rundensteiner, Jianjun Chen.* In 2026 IEEE International Conference on Data Engineering (ICDE 2026, Accepted).
5. **Agree to Disagree: Robust Anomaly Detection with Noisy Labels.** *Dennis M Hofmann, Peter M VanNostrand, Lei Ma, Huayi Zhang, Joshua C DeOliveira, Lei Cao, Elke A Rundensteiner.* Proceedings of the ACM on Management of Data (SIGMOD 2025).
6. **Pluto: Sample Selection for Robust Anomaly Detection on Polluted Log Data.** *Lei Ma, Lei Cao, Peter M. VanNostrand, Dennis M. Hofmann, Elke A. Rundensteiner.* Proceedings of the ACM on Management of Data 2025 (SIGMOD 2025).

7. **Reproducibility Report for ACM SIGMOD 2024 Paper: "FACET: Robust Counterfactual Explanation Analytics"**. *Peter M. VanNostrand, Konstantinos Kanellis, Wan Shen Lim, and Donghyun Sohn*. In Reproducibility Reports of the 2024 International Conference on Management of Data (SIGMOD ARI 2025).
8. **Counterfactual Explanation Analytics: Empowering Lay Users to Take Action Against Consequential Automated Decisions**. *Peter M. VanNostrand, Dennis M. Hofmann, Lei Ma, Belisha Genin, Randy Huang, and Elke A. Rundensteiner*. Proceedings of the VLDB Endowment (VLDB 2024).
9. **Actionable Recourse for Automated Decisions: Examining the Effects of Counterfactual Explanation Type and Presentation on Lay User Understanding**. *Peter M. VanNostrand, Dennis M. Hofmann, Lei Ma, Elke A. Rundensteiner*. Proceedings of ACM Conference on Fairness, Accountability, and Transparency (FAccT 2024).
10. **FACET: Robust Counterfactual Explanation Analytics**. *Peter M. VanNostrand, Huayi Zhang, Dennis M. Hofmann, Elke A. Rundensteiner*. Proceedings of the ACM on Management of Data 2023 (SIGMOD 2023).
11. **A Demonstration of AutoOD: a Self-Tuning Anomaly Detection System**. *Dennis Hofmann, Peter VanNostrand, Huayi Zhang, Yizhou Yan, Lei Cao, Samuel Madden, Elke Rundensteiner*. Proceedings of the VLDB Endowment 2022 (VLDB 2022).
12. **ELITE: Robust Deep Anomaly Detection with Meta Gradient**. *Huayi Zhang, Lei Cao, Peter VanNostrand, Samuel Madden, Elke A. Rundensteiner*. Proceedings ACM SIGKDD Conference on Knowledge Discovery & Data (KDD 2021).
13. **Confidential Deep Learning: Executing Proprietary Models on Untrusted Devices**. *Peter M. VanNostrand, Ioannis Kyriazis, Michelle Cheng, Tian Guo, Robert J. Walls*. arXiv:1908.10730 2019, Great Lakes Security Day 2019.

Presentations

1. **FACET: Finding Actionable Explanations for Unwanted AI Predictions**. *Peter M. VanNostrand, Huayi Zhang, Dennis Hofmann, and Elke A. Rundensteiner*. WPI Graduate Research Innovation Exchange, 2023. (Poster Presentation)
2. **Leveraging Labeled Examples for Robust Deep Anomaly Detection**. *Peter VanNostrand, Huayi Zhang, Lei Cao, Samuel Madden, and Elke Rundensteiner*. WPI Graduate Research Innovation Exchange, 2021. (Poster Presentation)
3. **Confidential Deep Learning: Executing Proprietary Models on Untrusted Devices**. *Peter M. VanNostrand, Ioannis Kyriazis, Michelle Cheng, Tian Guo, Robert J. Walls*. Great Lakes Security Day, 2019. (Research Talk)
4. **Optimized Simultaneous Localization and Mapping for Resource Limited Systems**. *Peter VanNostrand, Karthik Dantu*. SUNY Undergraduate Research Conference, 2019. (Poster Presentation)

Teaching and Mentorship

- Project Mentor, WPI Major Qualifying Project – Explainable AI**

Aug 2023 – May 2024

Guided a team of five undergraduate students in their first research project focusing on human-computer interfaces and design to create a lay user appropriate explanation UI
- Project Mentor, WPI Major Qualifying Project – Anomaly Detection**

Aug 2023 – May 2024

Guided a team of four undergraduate students in their first research project to develop a deployable web platform for domain experts to leverage deep anomaly detection methods
- Guest Lecturer, WPI NSF Research Experience for Undergraduates**

Summers 2020-2023

Served as a guest lecturer for the REU site *Interdisciplinary Data Science Research for Healthy Communities in the Digital Age* instructing 20+ students annually on best research practices and effective communication and presentation

Professional Experience

- Graduate Research Assistant**, Worcester Polytechnic Institute

Aug 2020 – Present
- Undergraduate Research Assistant**, Worcester Polytechnic Institute

May 2019 – Sep 2019
- Undergraduate Research Assistant**, University at Buffalo

Jun 2018 – May 2019
- Full Stack Web Developer**, Software Solutions Inc

May 2017 – Aug 2017

Honors and Awards

- FAcCT Travel Award**, Association of Computer Machinery

Jun 2024
- CEDAR PhD Fellowship** (2 years Tuition/Spend), National Science Foundation

Aug 2023
- People’s Choice Poster** (3rd Place), Graduate Research Innov. Exchange, WPI

Apr 2023
- Best Poster Runner Up**, Graduate Research Innovation Exchange, WPI

Apr 2022
- Community-Building Award**, WPI Data Science Department

May 2022
- Community-Building Award**, WPI Data Science Department

May 2021
- GAANN PhD Fellowship** (3 years Tuition/Spend), U.S. Dept of Education

Aug 2020
- Eagle Scout**, Boy Scouts of America

Aug 2016

Skills and Tools

Skills: Crowd sourced experimentation, A/B testing, dashboard design, data visualization, algorithmic auditing, technical/grant writing, project management, communication, policy

Tools: Python (Pytorch, Numpy, Pandas, Scikit-Learn, Jupyter), Javascript (React, D3, Node, JQuery), SQL, Java, C++, Git, Bash, LaTeX, MS Office, Adobe Illustrator, Davinci Resolve

References

Available upon request